# Exploratory Data Analysis: Covariation

PSY 410: Data Science for Psychology

Dr. Sara Weston

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## From variation to covariation

## Psychology is about relationships

Last time, you explored how individual variables behave — distributions, outliers, missing data.

But psychology is about *relationships*:

- Does **treatment** predict **depression**?
- Does **age** relate to **reaction time**?
- Does **anxiety** co-occur with **insomnia**?

Today we learn to **see** those relationships in data — before testing them statistically.

## Categorical + Continuous

## Example: Mental health by treatment group

```r
# Simulated therapy outcome data
therapy_data <- tibble(
  condition = rep(c("Control", "CBT", "Mindfulness"), each = 50),
  depression_post = c(
    rnorm(50, mean = 18, sd = 5),  # Control
    rnorm(50, mean = 12, sd = 5),  # CBT
    rnorm(50, mean = 14, sd = 5)   # Mindfulness
  )
)
```

## Example: Mental health by treatment group

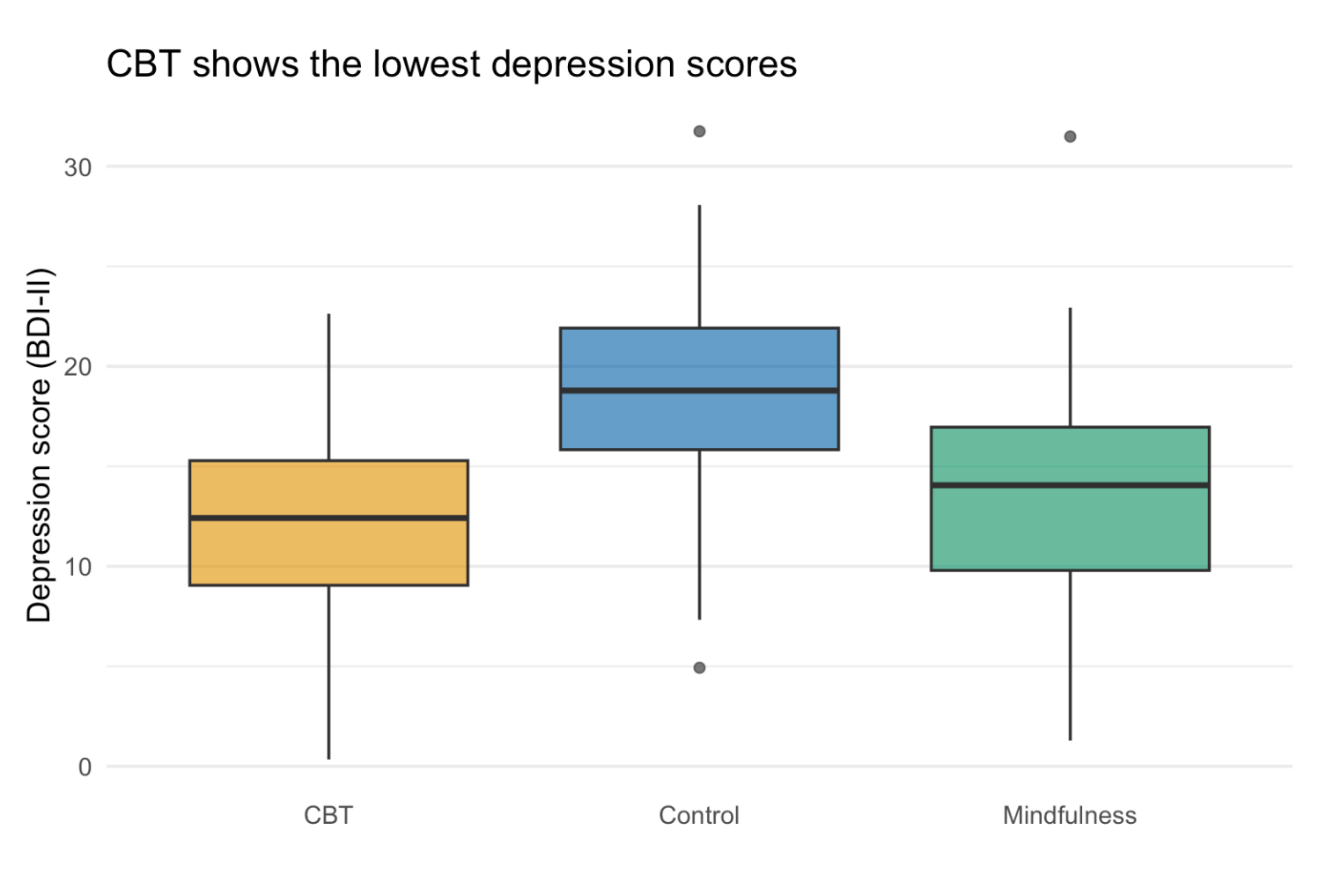
```r
glimpse(therapy_data)
```

```
Rows: 150
Columns: 2
$ condition       <chr> "Control", "Control", "Control", "Control",
"Control",…
$ depression_post <dbl> 26.559783, 18.316721, 13.498250, 28.065713,
16.112715,…
```

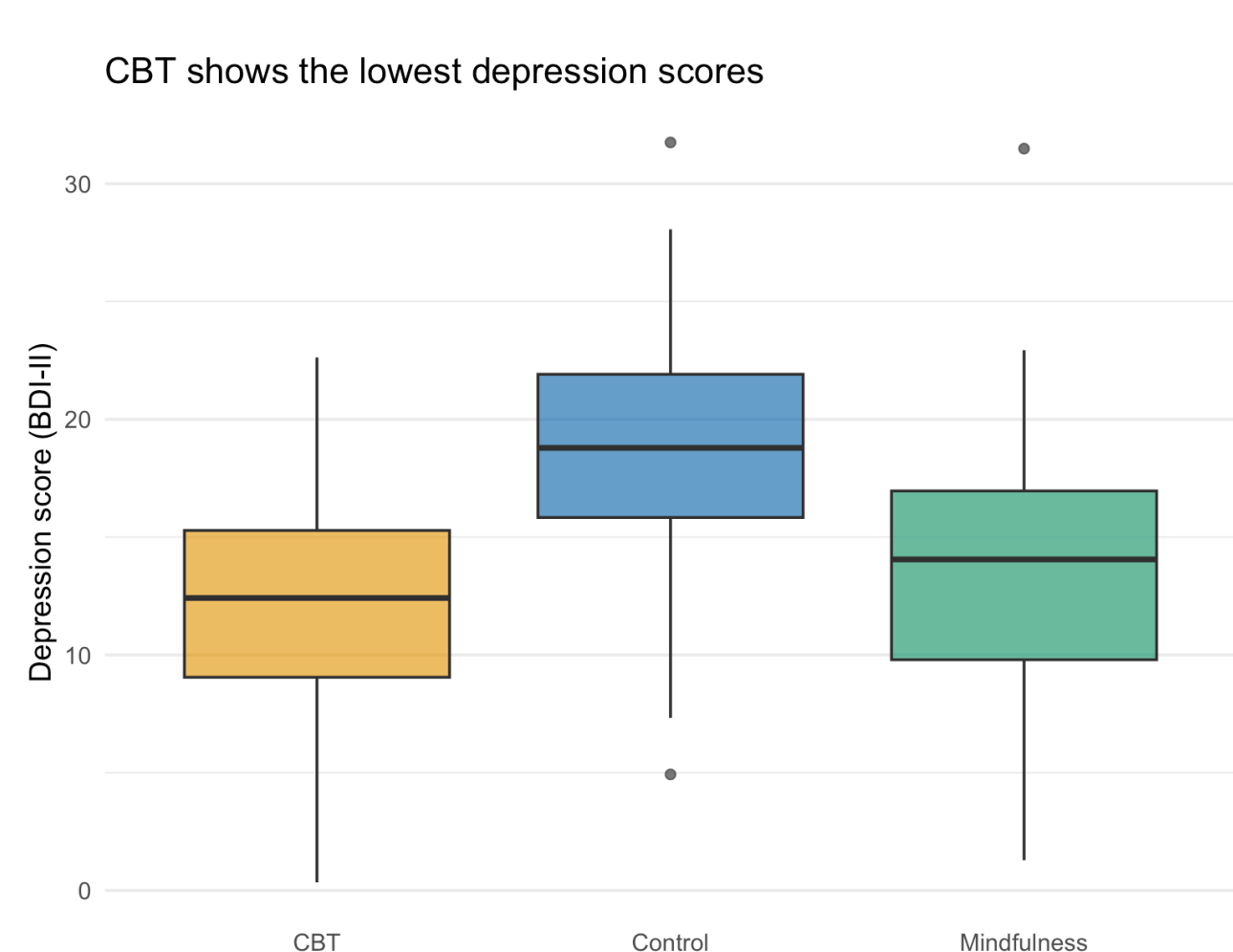
## Boxplots: The classic choice

```r
ggplot(therapy_data,
       aes(x = condition, y = depression_post, fill = condition)) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  scale_fill_manual(values = c(
    "Control" = "#0072B2", "CBT" = "#E69F00", "Mindfulness" = "#009E73")) +
  labs(
    title = "CBT shows the lowest depression scores",
    x = NULL,
    y = "Depression score (BDI-II)"
  ) +
  theme_minimal(base_size = 14) +
  theme(panel.grid.major.x = element_blank())
```

## Boxplots: The classic choice



## What boxplots show



- **Line in middle:** median
- **Box:** 25th–75th percentile (IQR)
- **Whiskers:** 1.5 × IQR
- **Dots:** outliers

…

Great for comparison — but they hide the actual data.

## The problem with boxplots

Boxplots summarize, but they hide important information:

- **The actual distribution shape** (is it bimodal? skewed?)
- **Individual data points** (how many observations are there?)
- **The raw data** (where do specific values fall?)

We can do better.

## Raincloud plots

## The modern psych visualization

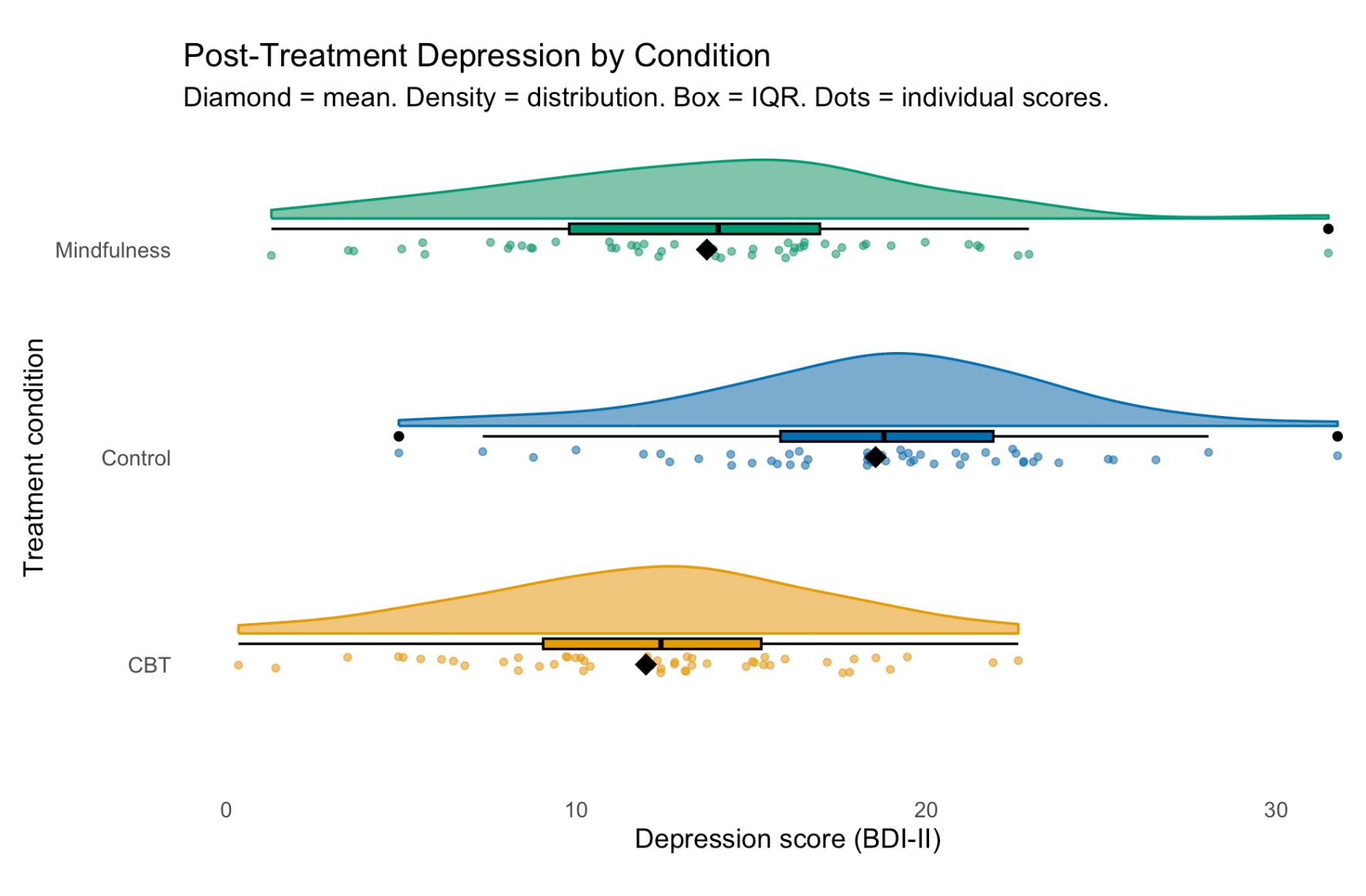
Raincloud plots combine three elements:

1. **Density curve** (distribution shape)
2. **Boxplot** (summary stats)
3. **Jittered points** (individual data)

They're increasingly popular in psychology publications because they show **everything**.

Use the `ggrain` package. (More details here.)

## Building a raincloud



```r
library(ggrain)
therapy_data |>
  ggplot(aes(
    x = condition,
    y = depression_post,
    fill = condition,
    color = condition)) +
  # geom_rain creates all parts of your raincloud
  geom_rain(
    alpha = .6,
    # change just the boxplot part
    boxplot.args = list(color = "black")) +
  # The mean
  stat_summary(fun = mean, geom = "point", shape = 18, size = 5, color = "bla
  scale_fill_manual(
```

## When to use which

### Boxplot

- Median, IQR, outliers
- Fast to read
- Best for **quick EDA** and large datasets where individual points would clutter

### Raincloud

- Density + summary + raw data
- Shows shape, sample size, and outliers all at once
- Best for **publications and presentations**

## Pair coding break

## Your turn: Build a raincloud

Using `therapy_data`, build a **raincloud plot** of depression scores by condition.

1. Load `ggrain`
2. Map `x = condition`, `y = depression_post`, and `fill = condition`
3. Add `geom_rain()`
4. Give it an informative title and labeled axes

**Time: 10 minutes**

## Before we move on

📤 Submit your code on Canvas for participation credit. Paste what you have — it doesn't need to work perfectly.

## Categorical + Categorical

# Example: Diagnosis by referral source

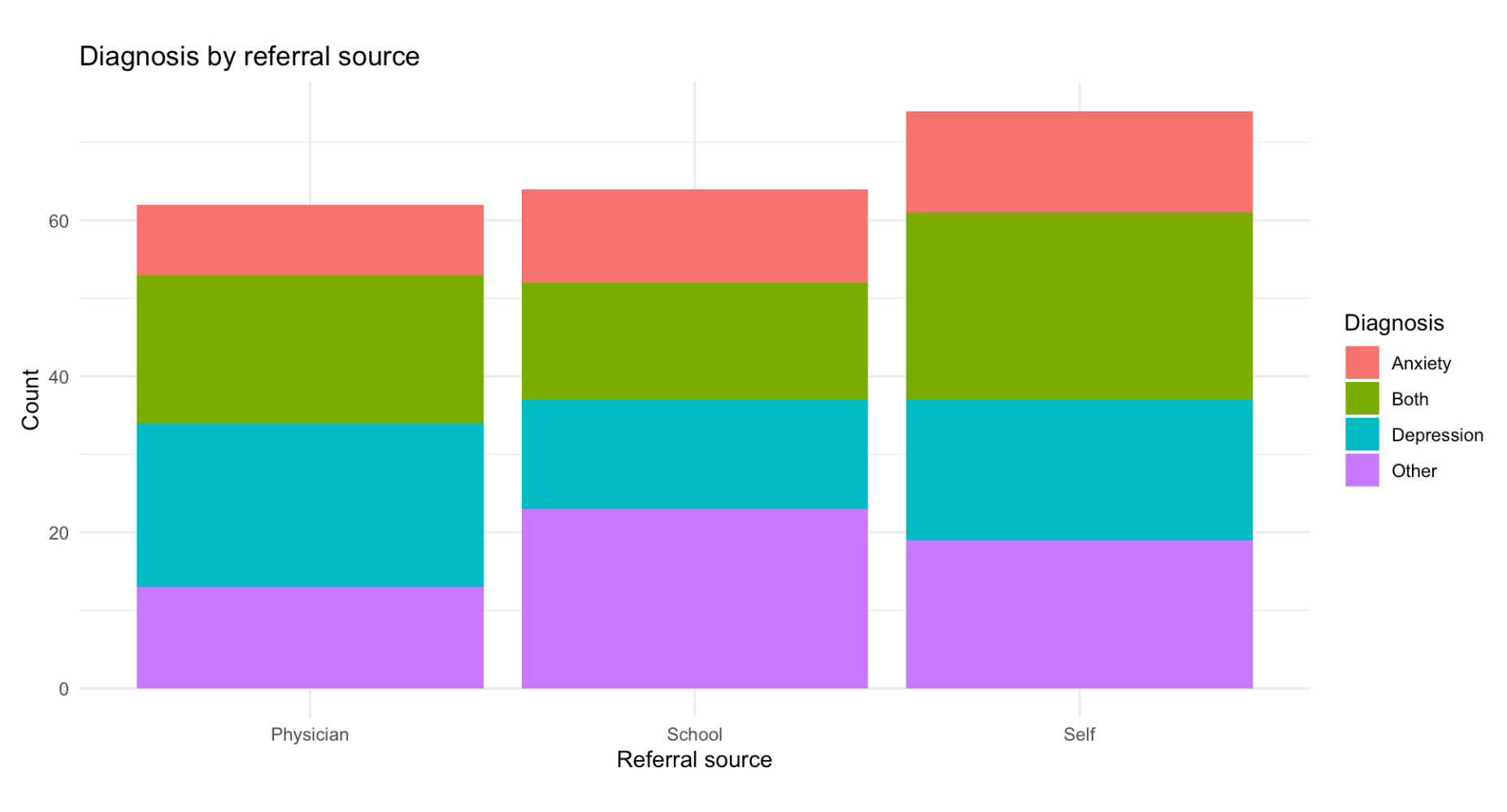
```r
# Simulated diagnostic data
diagnosis_data <- tibble(
  referral = sample(c("Self", "Physician", "School"), 200, replace = TRUE),
  diagnosis = sample(c("Depression", "Anxiety", "Both", "Other"),
                     200, replace = TRUE)
)

head(diagnosis_data)
```

```
# A tibble: 6 × 2
  referral  diagnosis
  <chr>     <chr>
1 Self      Anxiety
2 Physician Depression
3 Self      Other
4 Self      Depression
5 Physician Both
6 School    Other
```

# Option 1: Stacked bar chart

```r
ggplot(diagnosis_data, aes(x = referral, fill = diagnosis)) +
  geom_bar() +
  labs(
    title = "Diagnosis by referral source",
    x = "Referral source",
    y = "Count",
    fill = "Diagnosis"
  ) +
  theme_minimal()
```

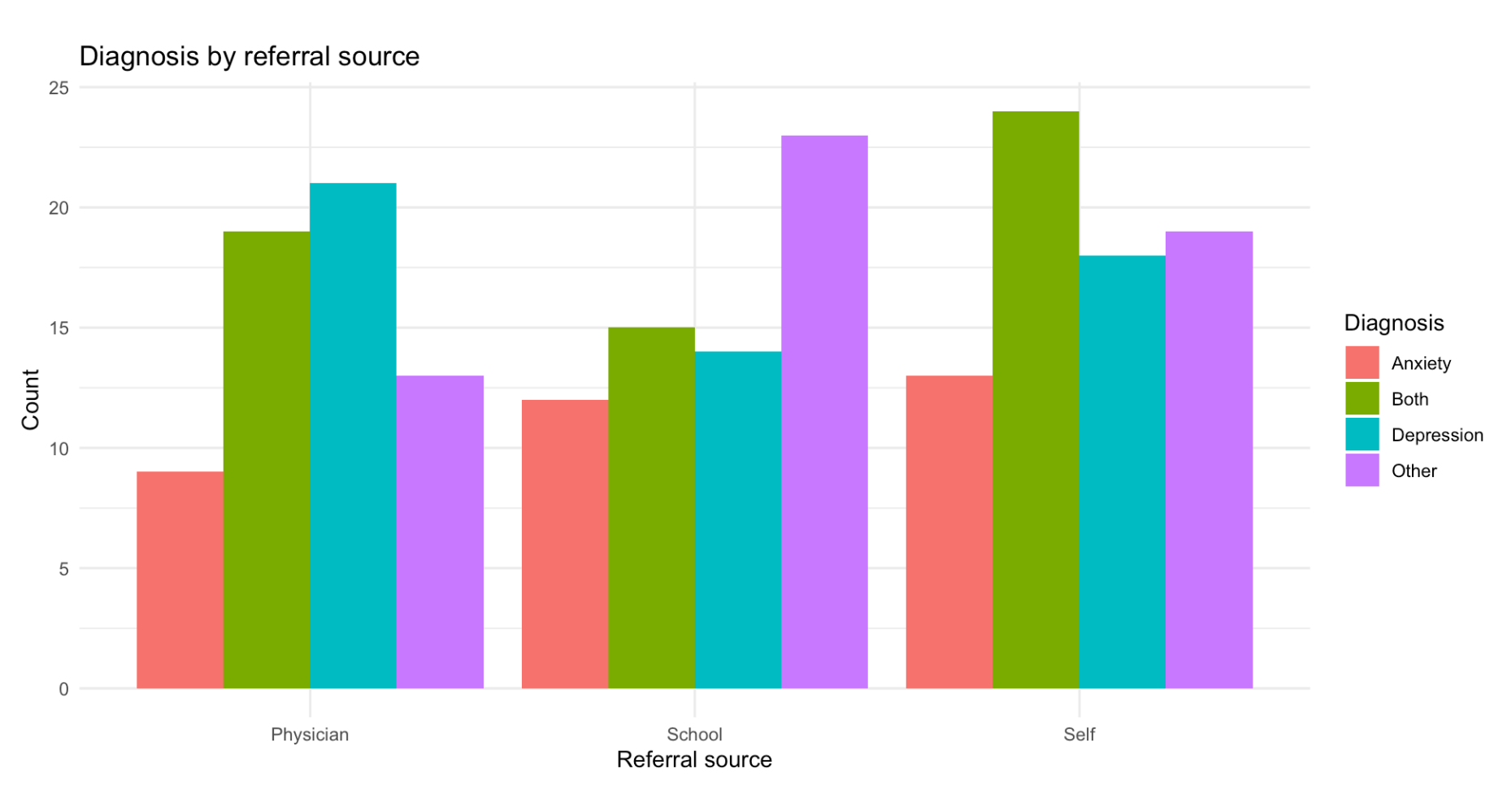
# Option 1: Stacked bar chart



# Option 2: Side-by-side bars

```r
ggplot(diagnosis_data, aes(x = referral, fill = diagnosis)) +
  geom_bar(position = "dodge") +
  labs(
    title = "Diagnosis by referral source",
    x = "Referral source",
    y = "Count",
    fill = "Diagnosis"
  ) +
  theme_minimal()
```

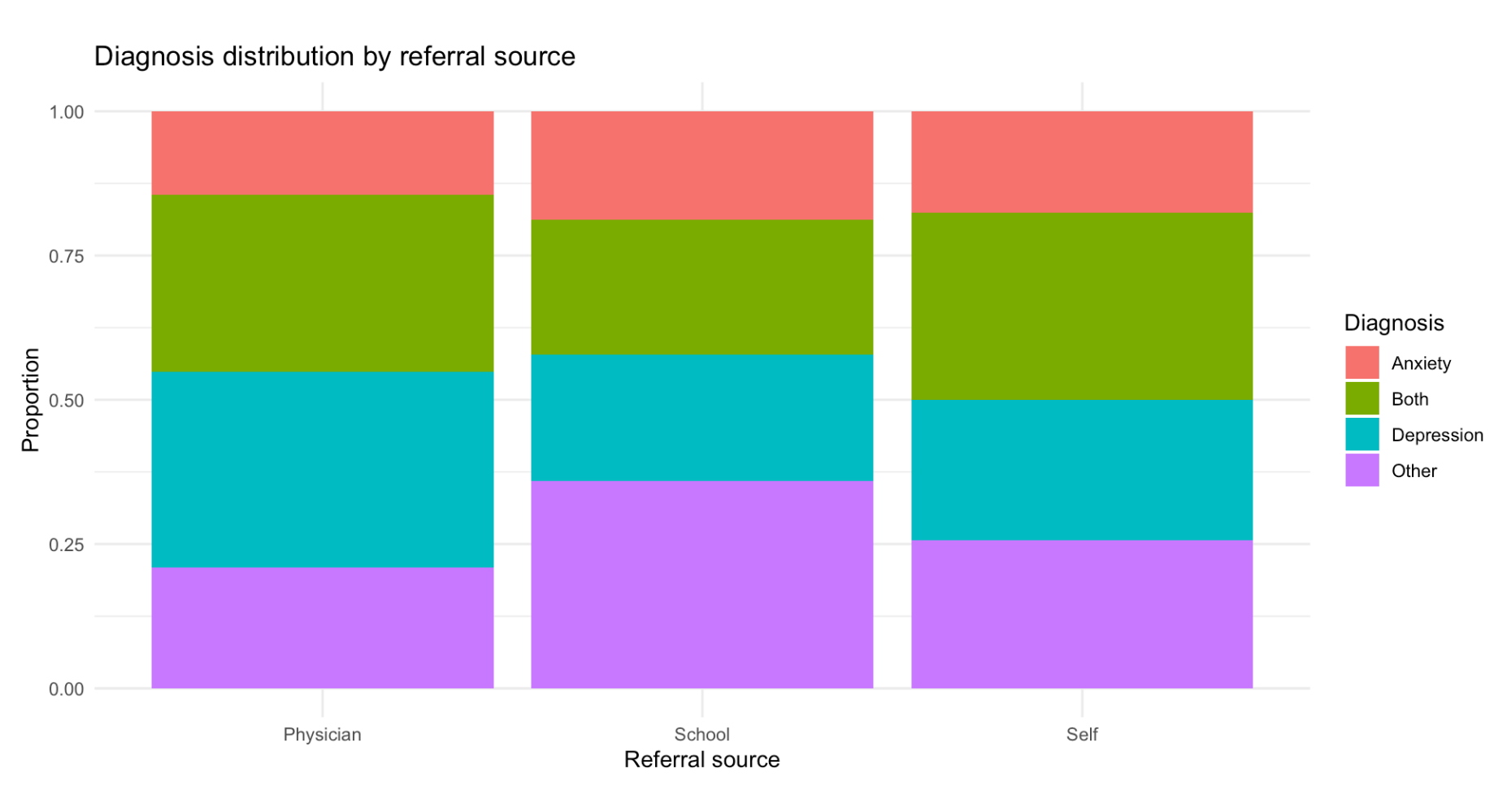
# Option 2: Side-by-side bars



# Option 3: Proportions

```r
ggplot(diagnosis_data, aes(x = referral, fill = diagnosis)) +
  geom_bar(position = "fill") +
  labs(
    title = "Diagnosis distribution by referral source",
    x = "Referral source",
    y = "Proportion",
    fill = "Diagnosis"
  ) +
  theme_minimal()
```

# Option 3: Proportions



# Continuous + Continuous

# The classic: Scatterplots

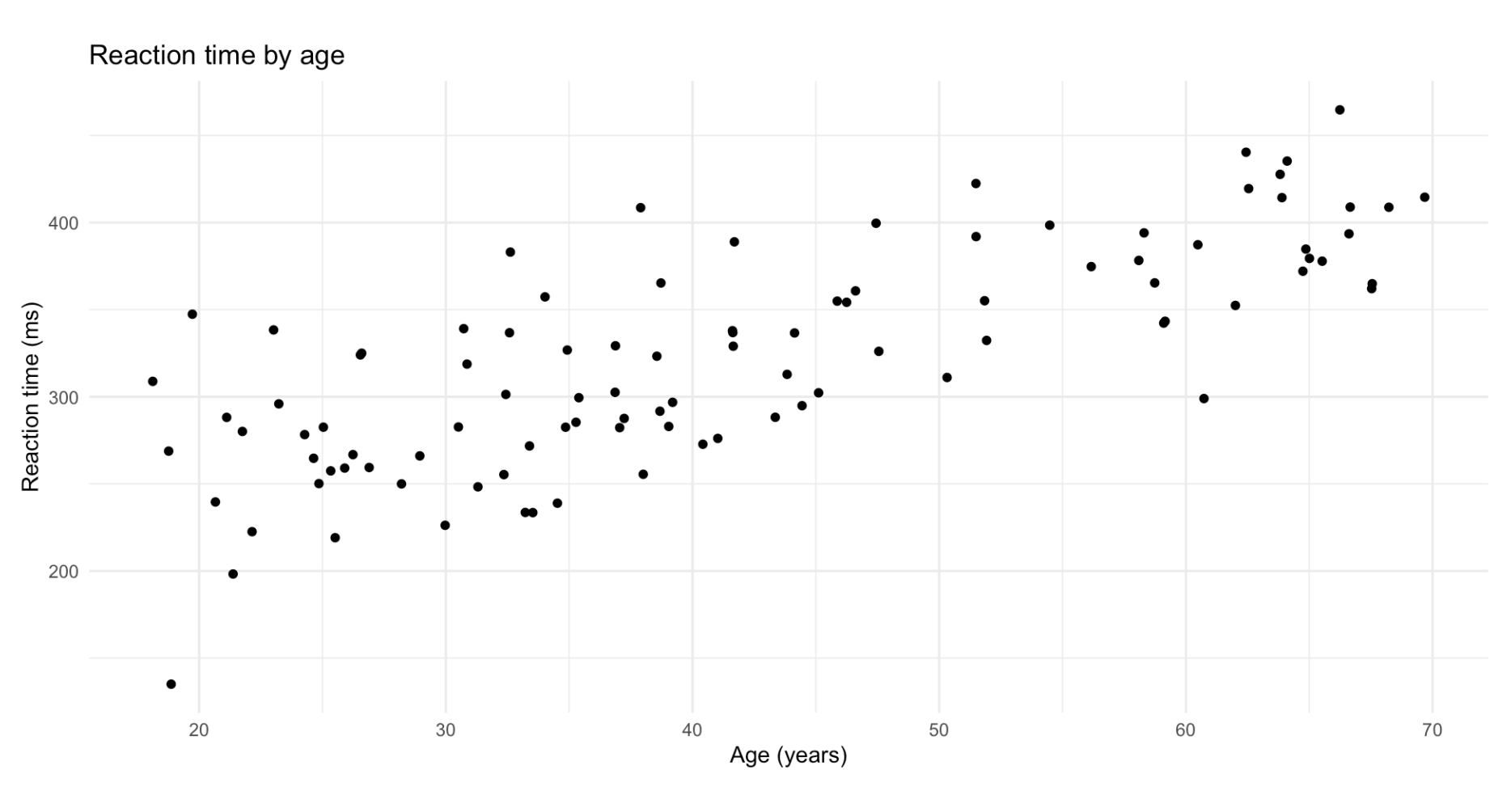
```r
# Simulated reaction time data
rt_data <- tibble(
  age = runif(100, 18, 70),
  reaction_time = 200 + age * 3 + rnorm(100, 0, 40)
)

glimpse(rt_data)
```

```
Rows: 100
Columns: 2
$ age           <dbl> 38.00245, 37.89727, 67.53066, 63.82036, 18.76507,
47.552…
$ reaction_time <dbl> 255.5022, 408.5319, 362.0651, 427.6740, 268.7969,
326.06…
```

# Basic scatterplot

```r
ggplot(rt_data, aes(x = age, y = reaction_time)) +
  geom_point() +
  labs(
    title = "Reaction time by age",
    x = "Age (years)",
    y = "Reaction time (ms)"
  ) +
  theme_minimal()
```

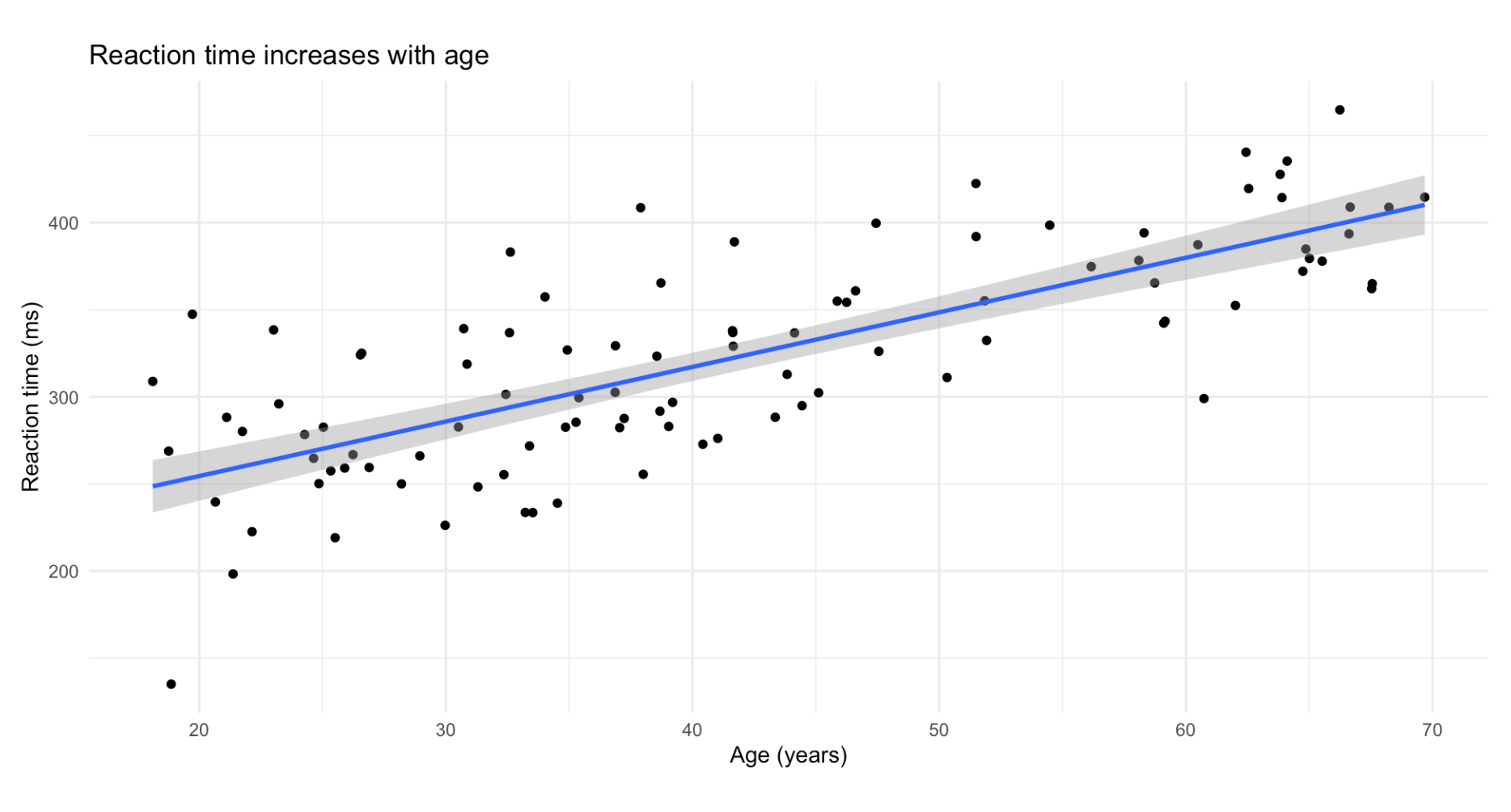
# Basic scatterplot



# Add a trend line

```r
ggplot(rt_data, aes(x = age, y = reaction_time)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Reaction time increases with age",
    x = "Age (years)",
    y = "Reaction time (ms)"
  ) +
  theme_minimal()
```

# Add a trend line



# The problem: Overplotting

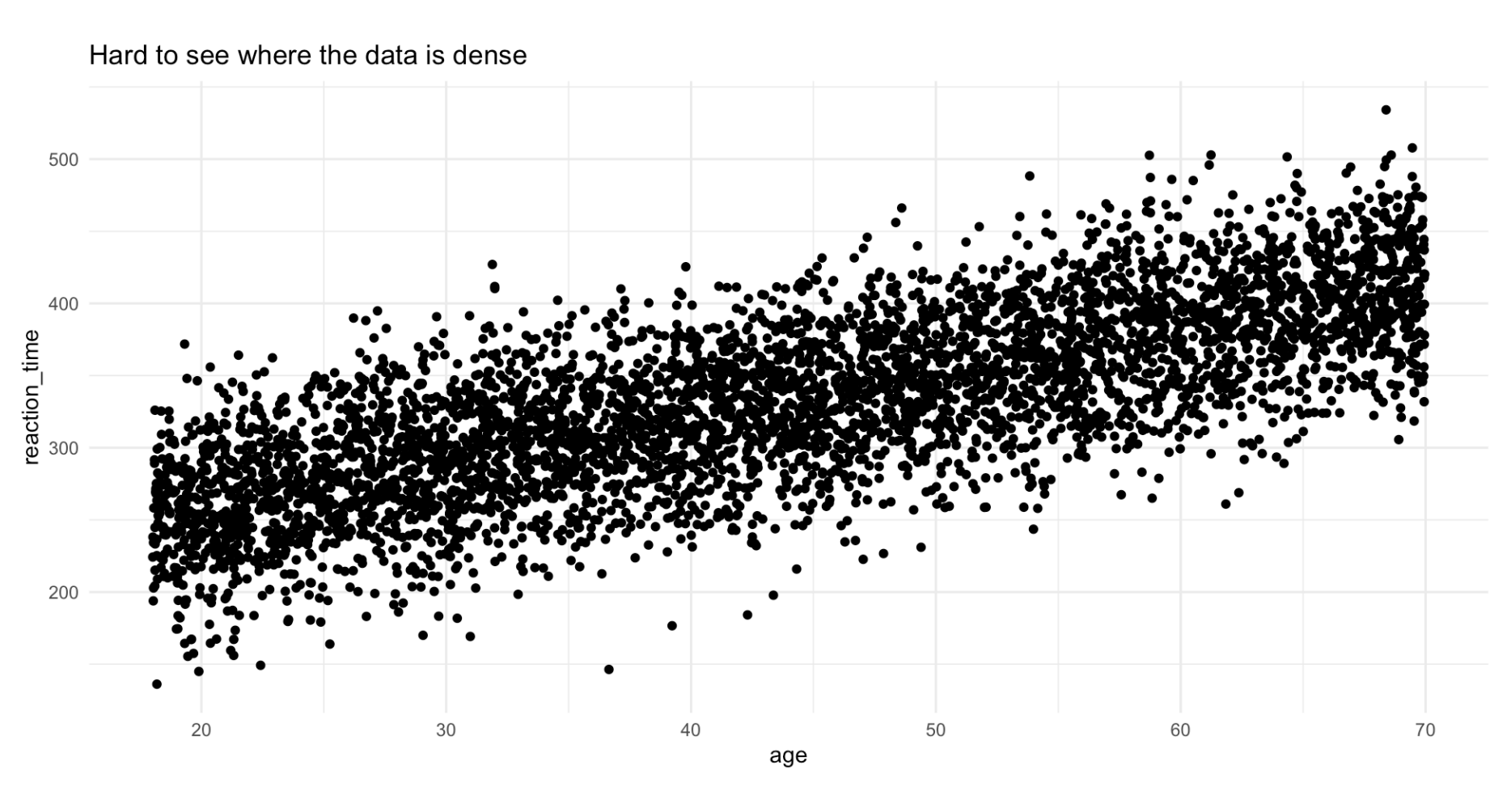
With lots of data, points overlap and hide the true density:

```r
# Lots of data
big_rt_data <- tibble(
  age = runif(5000, 18, 70),
  reaction_time = 200 + age * 3 + rnorm(5000, 0, 40)
)
```

# Overplotting problem demonstrated

```r
ggplot(big_rt_data, aes(x = age, y = reaction_time)) +
  geom_point() +
  labs(title = "Hard to see where the data is dense") +
  theme_minimal()
```

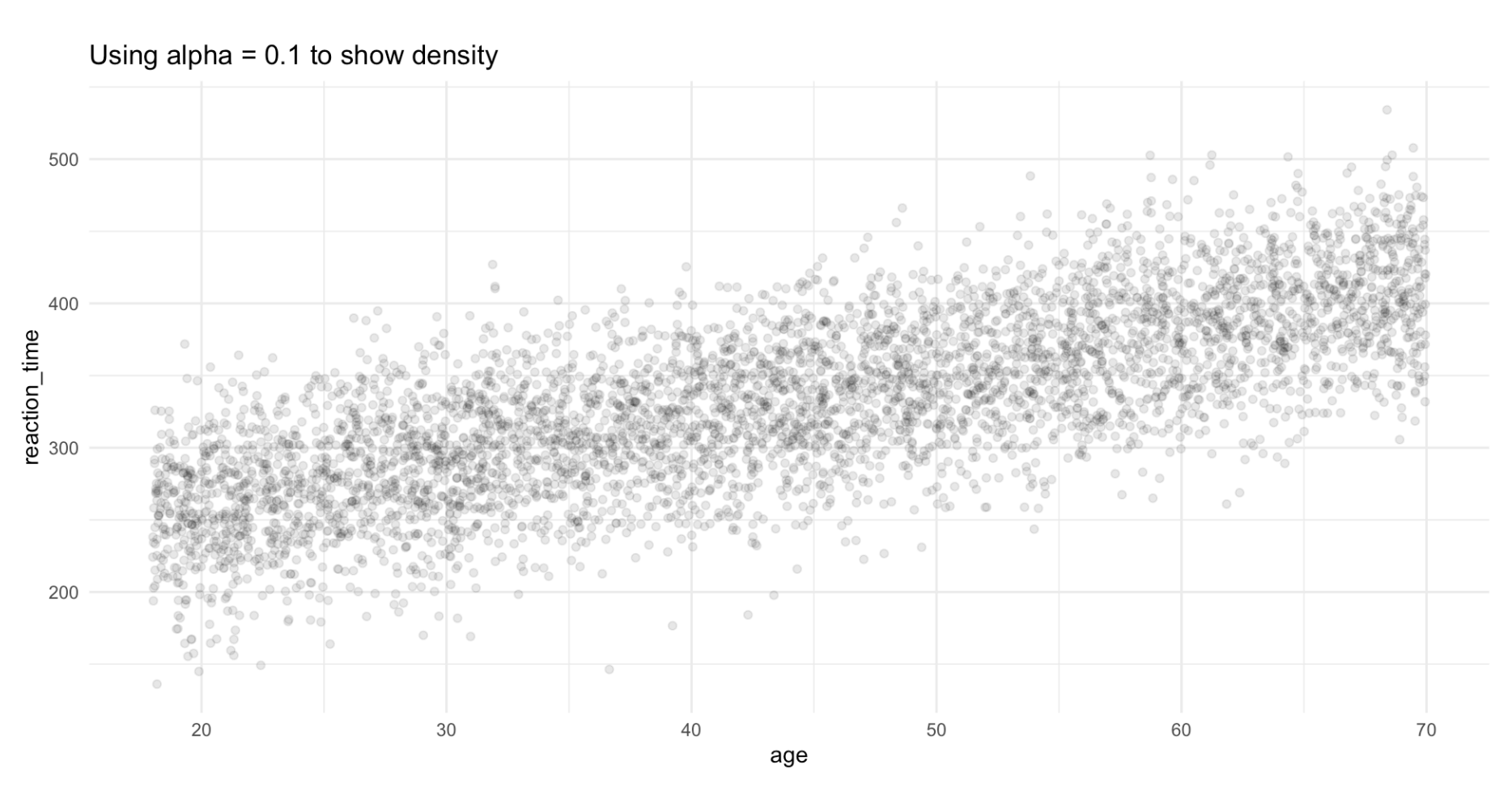
# Overplotting problem demonstrated



# Solution 1: Transparency (alpha)

```r
ggplot(big_rt_data, aes(x = age, y = reaction_time)) +
  geom_point(alpha = 0.1) +
  labs(title = "Using alpha = 0.1 to show density") +
  theme_minimal()
```

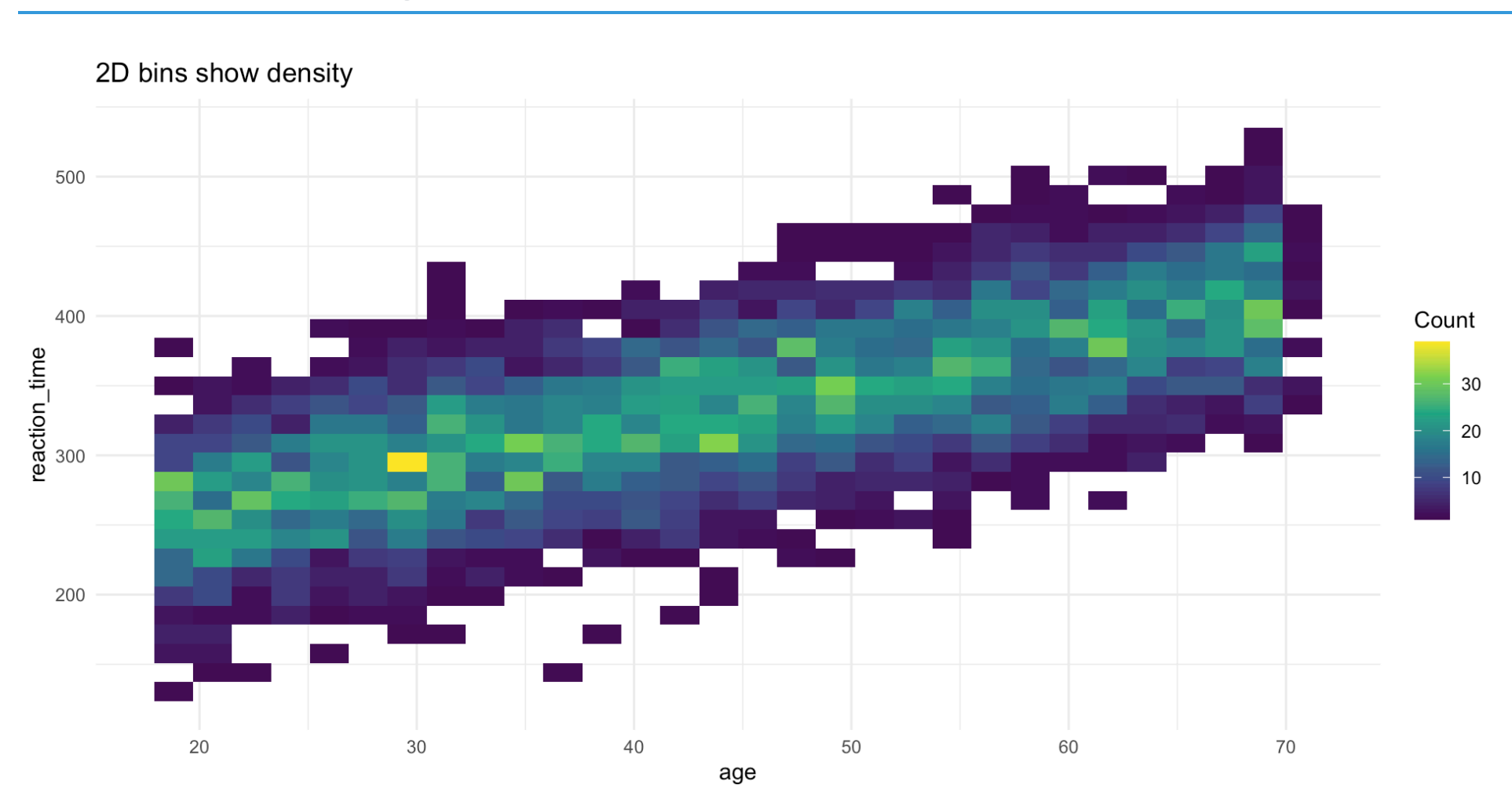
# Solution 1: Transparency (alpha)



# Solution 2: geom_bin2d()

```r
ggplot(big_rt_data, aes(x = age, y = reaction_time)) +
  geom_bin2d() +
  scale_fill_viridis_c() +
  labs(
    title = "2D bins show density",
    fill = "Count"
  ) +
  theme_minimal()
```

## Solution 2: geom_bin2d()



## Correlation coefficient

A single number summary of the linear relationship:

```
cor(rt_data$age, rt_data$reaction_time)
```

```
[1] 0.7604132
```

ⓘ Note

- **r = 1:** perfect positive relationship
- **r = 0:** no linear relationship
- **r = -1:** perfect negative relationship

But always plot your data first! (See: Anscombe's Quartet)

# Patterns and models

## What patterns tell us

When you see covariation, ask:

1. **Could it be coincidence?** (Maybe, especially with small samples)
2. **What's the mechanism?** (How are these variables related?)
3. **Is there a confound?** (Could a third variable explain both?)

⚠ Warning

**Correlation ≠ Causation**

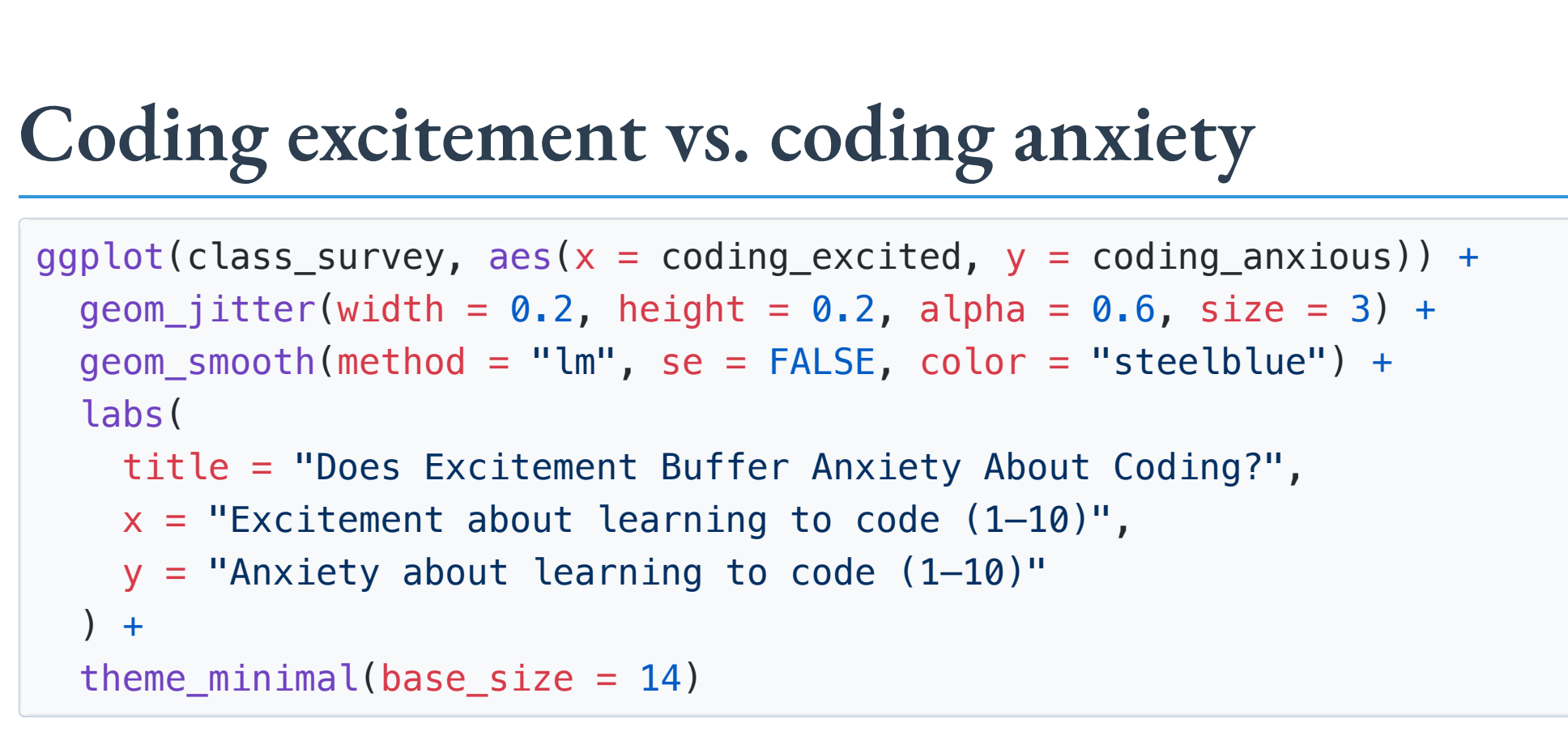
Covariation suggests a relationship, but doesn't prove one variable *causes* the other.

# Your class data: Covariation

## Stress by chronotype

```
ggplot(class_survey, aes(x = chronotype, y = stress, fill = chronotype)) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  labs(
    title = "Are Night Owls More Stressed?",
    x = "Chronotype",
    y = "Current stress level (1–10)"
  ) +
  theme_minimal(base_size = 14)
```

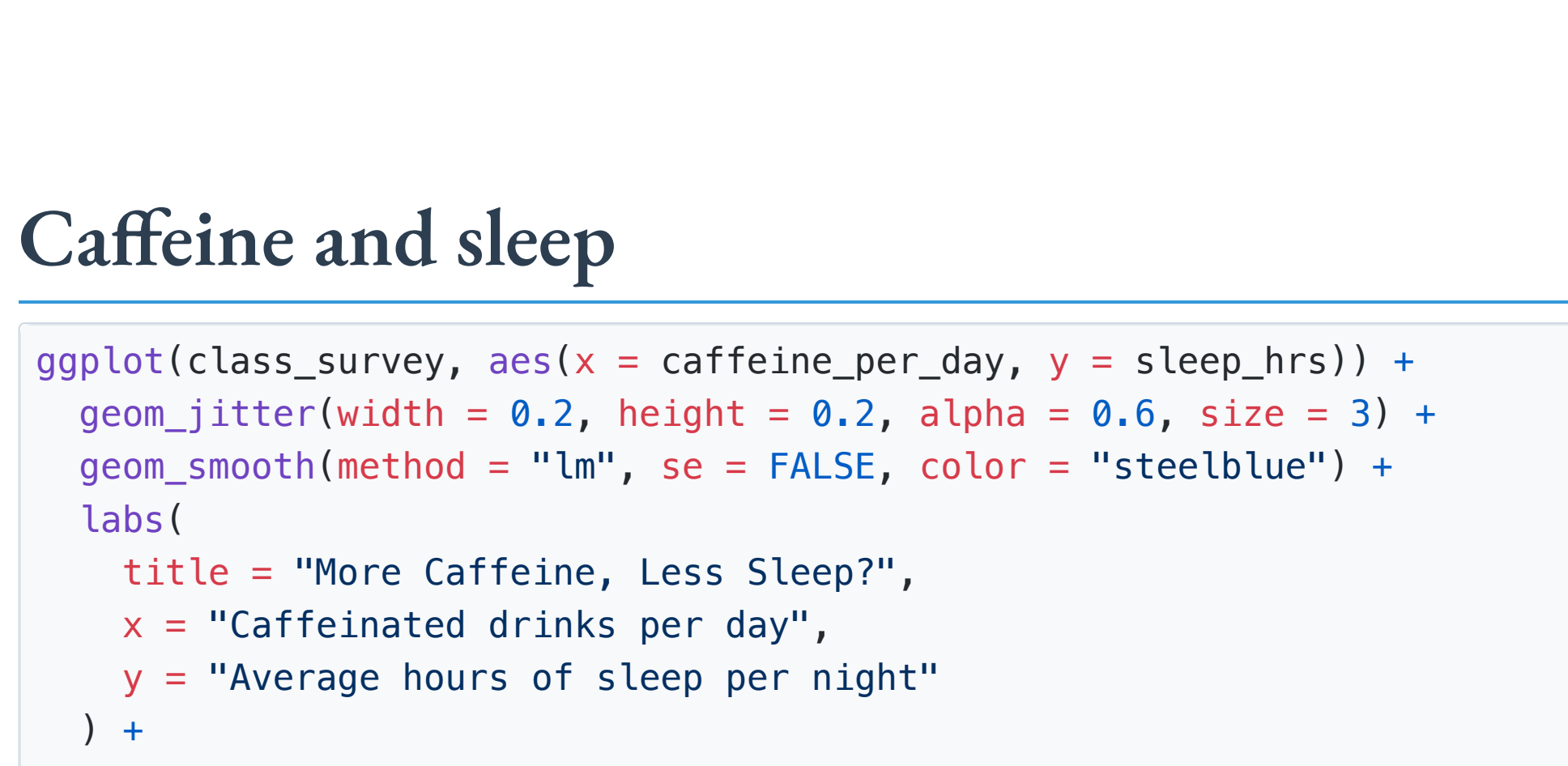
## Stress by chronotype



## Coding excitement vs. coding anxiety

```
ggplot(class_survey, aes(x = coding_excited, y = coding_anxious)) +
  geom_jitter(width = 0.2, height = 0.2, alpha = 0.6, size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "steelblue") +
  labs(
    title = "Does Excitement Buffer Anxiety About Coding?",
    x = "Excitement about learning to code (1–10)",
    y = "Anxiety about learning to code (1–10)"
  ) +
  theme_minimal(base_size = 14)
```

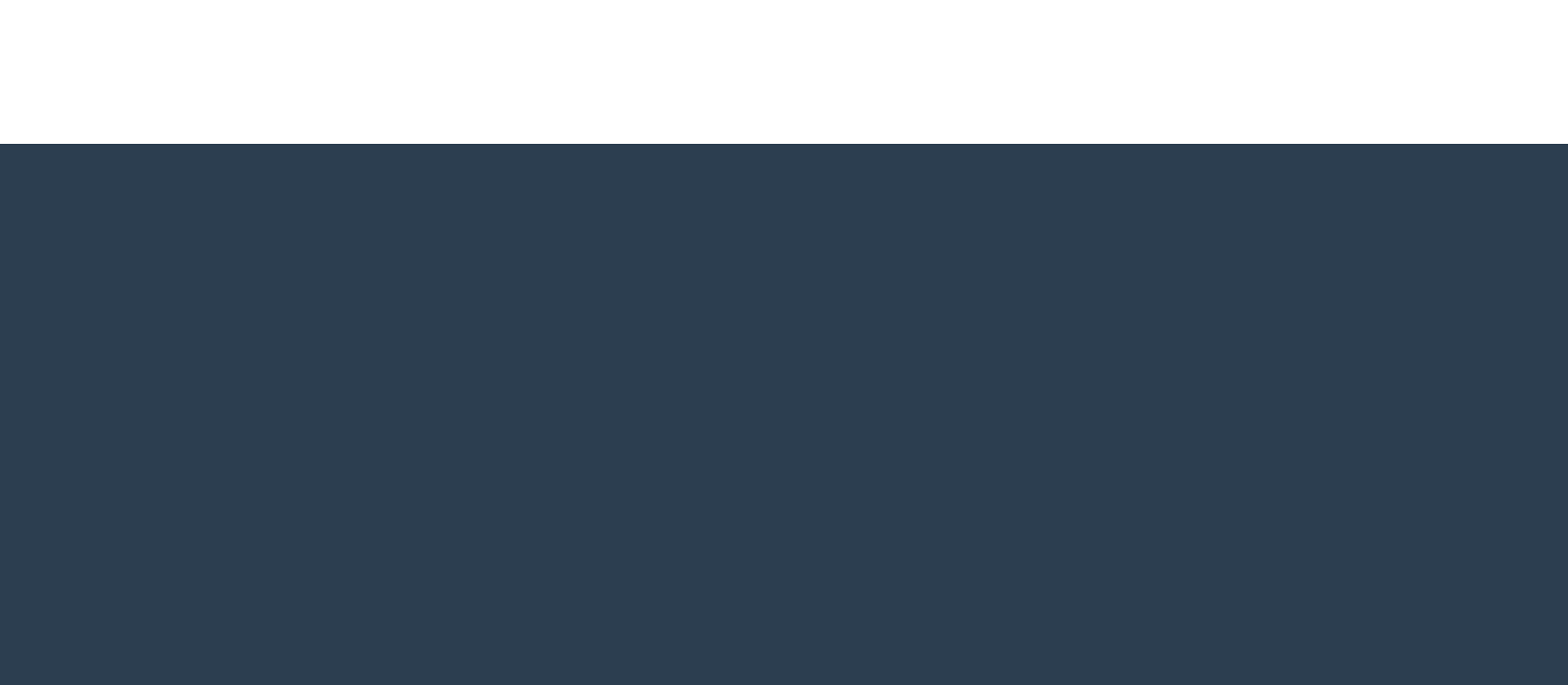
## Coding excitement vs. coding anxiety



## Caffeine and sleep

```
ggplot(class_survey, aes(x = caffeine_per_day, y = sleep_hrs)) +
  geom_jitter(width = 0.2, height = 0.2, alpha = 0.6, size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "steelblue") +
  labs(
    title = "More Caffeine, Less Sleep?",
    x = "Caffeinated drinks per day",
    y = "Average hours of sleep per night"
  ) +
  theme_minimal(base_size = 14)
```

## Caffeine and sleep



# Wrapping up

## Key takeaways

1. **Covariation = relationships** between variables
2. **Different plot types** for different variable combinations:
    - Categorical + continuous: boxplot, raincloud
    - Categorical + categorical: stacked, dodged, or proportional bars
    - Continuous + continuous: scatterplot (with alpha or 2D bins for big data)
3. **Rainclouds** show distribution, summary stats, and raw data all at once
4. **Watch for overplotting** — use alpha, jitter, or binning
5. **Always visualize first** before computing correlations
6. **Patterns suggest but don't prove** causation

## Before next class

📖 **Read:**

- R4DS Ch 12: Logical vectors
- R4DS Ch 13: Numbers

✅ **Do:**

- Get started on Assignment 5 (due Sun May 10)
- Start thinking about how you'll compute scale scores — that's where we're headed Wednesday

## The one thing to remember

Relationships hide in data. Your job is to make them visible — carefully, honestly.

See you Wednesday!

# Get a head start

## Assignment 5 preview

Assignment 5 asks you to do EDA on the **BFI** personality dataset. Two of the tasks line up directly with what we did today:

- **Task 2.2:** Scatterplot of two Extraversion items with a trend line
- **Task 2.3:** Boxplots of one Extraversion item by education level

If you have time tonight, take a first pass at either one — the code patterns are the same as today's class survey examples.